

CRF State Summary v9

K20–K24 Session — April 2026

Maturity: 8.0/8 (Theory) · 7.5/8 (Empirical gate open)

Ougit Jarittum · ORCID: 0009-0009-4451-6811

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Abstract

This session discovers five new identities (K20–K24) via pattern mining. The most important: K20 ($K^*T_{\text{avg}}/F = e$, gap 0.005%) introduces Euler’s number into CRF and derives F from $(D_0, n, \varepsilon_0, e)$ alone. K21 ($T_{\text{avg}}\sigma_\Psi/\varphi_{\text{var}} = \sqrt{\varphi}$, gap 0.004%) is the sharpest identity in the corpus. Both K20 and K21 are *specific to* $n = 8$, checked across $n = 3 \dots 12$. Combined with K11 ($n = 2^{d_s}$) and K21, $n = 8$ is now pinned by four independent mathematical conditions. The gap $\beta_{\text{K11}}/\beta_0 = 2 \ln^2 2/D_0 = 1.022$ is identified as the irreducible finite- n correction; $\beta = d_s/(2 \ln 2)$ is exact only at $n \rightarrow \infty$.

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1 Primitive Chain

$$\delta \rightarrow \mathcal{P} \rightarrow \mathcal{C} \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow \mathcal{R} \rightarrow t$$

With $n = 2^{d_s}$: $\ln n = d_s \ln 2 = d_s \alpha$. Master chain: $\delta \rightarrow n \rightarrow D_0 = n(1 - e^{-1/n}) \rightarrow \beta = \ln n/D_0$, $\kappa = \sqrt{d_s} \alpha/n$.

2 New This Session: K20–K24

ID	Identity	Gap	Significance
K20	$K^*T_{\text{avg}}/F = e$	0.005%	Derives F ; e enters CRF
K21	$T_{\text{avg}}\sigma_{\Psi}/\varphi_{\text{var}} = \sqrt{\varphi}$	0.004%	Sharpest identity; $n = 8$ pinned
K22	$H_u\beta/H_b = e$	0.041%	Derives H_u from K3+K11
K23	$H_a\beta/\varphi = \sqrt{2d_s}$	0.044%	$\sqrt{6}$ web
K24	$n\psi_{\text{grad}}/K^* = \sqrt{6}$	0.046%	$\sqrt{6}$ web
<i>Structural: $n = 8$ is pinned by four independent conditions</i>			
K11	$n = 2^{d_s}$ (binary Born)	exact	Prerequisite for K11
K20	$K^*T_{\text{avg}}/F = e$ at $n = 8$ only	0.005%	Fails all other n
K21	$T_{\text{avg}}\sigma_{\Psi}/\varphi_{\text{var}} = \sqrt{\varphi}$ at $n = 8$ only	0.004%	Fails all other n
K11	$\beta D_0 = \ln n$	0.005%	Structural

3 Complete Status Table

Result	Status	Note
<i>Foundations</i>		
$n = 3$ from $\text{SO}(3)$	Derived	#16
$\alpha = \ln 2$	Derived	Session 8
Schwarzschild, $G = \ln 2 \cdot c^2 / D_0$	Derived	Paper v4
<i>Resolution lattice</i>		
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$K^* = D_0/n$ (K1)	Derived	0.43%
$\theta(K^*) = K^*$	Derived	0.001%
$n = 2^{d_s}$	Physical arg	exact
<i>Information constants</i>		
$\beta \cdot D_0 = \ln n$ (K11)	Derived	0.005%
$\kappa = \sqrt{d_s} \ln 2 / n$ (K7)	Hypothesis	0.047%
$\kappa / K^* = \beta / \sqrt{d_s}$ (K17)	Derived	exact
$\beta / \kappa = n \sqrt{d_s} / D_0$ (K18)	Derived	exact
$\beta \kappa = \ln^2 n / (n \sqrt{d_s} D_0)$ (K19)	Derived	0.005%
<i>e-web (new)</i>		
$K^* T_{\text{avg}} / F = e$ (K20)	Hypothesis	0.005%, new; $n = 8$ specific
$H_u \beta / H_b = e$ (K22)	Hypothesis	0.041%, new
<i>$\sqrt{6}$-web (new)</i>		
$H_a \beta / \varphi = \sqrt{2 d_s}$ (K23)	Hypothesis	0.044%, new
$n \psi_{\text{grad}} / K^* = \sqrt{6}$ (K24)	Hypothesis	0.046%, new
<i>φ-web</i>		
$T_{\text{avg}} \sigma_{\Psi} / \varphi_{\text{var}} = \sqrt{\varphi}$ (K21)	Hypothesis	0.004%, new; $n = 8$ specific
$m_{\text{fire}} = \varphi$ (K6)	⁴ Hypothesis	0.06–0.27%
$F \cdot \varphi^2 = 4$ (K14)	Hypothesis	0.057%

4 Key Numbers v9

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8 = 2^3$	$N = 500$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5378$
$T_{\text{avg}} = 35.57$	$\varphi = 1.61803$	$p = 9/10$	$H_a = 1.7925$	$H_b = 1.6717$
$\varphi_{\text{var}} = 0.4844$	$\sigma_\Psi = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$H_u = 2.055$	$e = 2.71828$

5 Open Gaps

Priority 1: $n = 2^{d_s}$ formal proof from δ -chain

The binary Born rule argument is physically motivated. Formal derivation: show that the δ -chain requires $n = 2^{d_s}$ Born-rule modes in d_s spatial dimensions. This would make K11, K17, K18, K20 (at $n = 8$) all exact from δ .

Priority 2: Physical proof of K20 ($F \cdot e = K^* T_{\text{avg}}$)

Show that the Born-resampling process in the $n = 8$ KL-neighbour graph satisfies $F \cdot e = 2D_0^2/(n(n-1)\varepsilon_0)$. The appearance of e suggests an exponential self-normalisation condition on the wave amplitude.

Priority 3: $m(m-1) = 1$ from energy balance

With K19: $\beta\kappa = \ln^2 n/(n\sqrt{d_s}D_0)$. The energy balance $m(m-1) = \beta\kappa/f(\delta)$ has $m = \varphi$ when $f = \beta\kappa$. Showing $f = \beta\kappa$ from first principles closes this gap.

6 Documents This Session

Document	Content
CRF_GoldenClosure.pdf	$\psi_{\text{grad}} \perp \Delta H$ (K3); $m = \varphi$ clarification
CRF_NewIdentities_K7K10.pdf	K7–K10; AR(1) from n
CRF_K20K24.pdf	K20–K24; e in CRF; $n = 8$ pinned
CRF_StateSummary_v9.pdf	This document

$$F \cdot e = K^* \cdot T_{\text{avg}}.$$

At $n = 8$: the wave, the threshold, the timing, and Euler's constant are one equation.

At $n = 9$: they are not. At $n = 7$: they are not.

$n = 8$ is not chosen. It is the only value where everything agrees.